

Deep Learning Detection of Beyond-General-Relativity Deviations in Gravitational-Wave Signals: A Detection-Threshold Study with Real LIGO Noise

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Abstract

We study machine-learning detection of controlled beyond-General-Relativity (beyond-GR) deviations in gravitational-wave signals, using both synthetic aLIGO-PSD noise and real LIGO H1 detector strain. Three deviation families are applied to General-Relativistic inspiral-merger-ringdown waveforms: amplitude modulation, phase modulation, and frequency modulation, each parameterized by a dimensionless strength coefficient β . A hybrid classifier combining a one-dimensional convolutional neural network with ten hand-crafted waveform statistics is trained on GR and modified waveforms and tested on a deviation type excluded from training. The central result is a quantitative detectability curve as a function of β . Using the real GW150914 strain as a template and real H1 detector noise, we find a detection threshold at $\beta \approx 0.25$, with accuracy rising smoothly from chance at $\beta \leq 0.2$ to perfect classification at $\beta \geq 0.5$. The threshold value is specific to the quadratic-in-time modulation form adopted here and should not be interpreted as a generic constraint on beyond-GR parameters. We nevertheless argue that the negative result at small β is informative: it establishes a quantitative limit on machine-learning-only beyond-GR searches in real detector noise, in the absence of matched-filter signal extraction.

1 Introduction

The direct detection of gravitational waves by the LIGO and Virgo observatories [1] has opened a new observational window onto strong-field gravity. Every detected event to date is consistent with the predictions of General Relativity (GR) to within measurement uncertainties [2]. Nevertheless, the extreme regimes probed by binary black hole mergers make gravitational waves a natural laboratory for testing the foundational assumptions of GR. Traditional tests rely on matched filtering against parametrized waveform families [3]. This approach is powerful when the deviation lies within the assumed template manifold, but it becomes computationally expensive in high-dimensional parameter spaces and cannot detect deviations that fall outside the template family. Machine-learning (ML) methods offer a complementary strategy: rather than matching against a specific template, they can learn discriminative features directly from data. Recent work has applied deep learning to gravitational-wave detection [4, 5], parameter estimation [6, 7], and glitch classification [8]. However, the application of ML to beyond-GR searches in real detector data has not been systematically characterized.

This paper addresses four questions. First, can a hybrid neural-network classifier distinguish GR waveforms from systematically modified (beyond-GR) waveforms in real LIGO detector noise? Second, what is the minimum deviation strength—parameterized by the amplitude-modulation coefficient β —that this classifier can reliably detect? Third, does the classifier

generalize to deviation types that were never presented during training? Fourth, how does classification performance depend on the signal-to-noise ratio of the injected signal, the deviation type, and the binary mass configuration?

We make five contributions. We construct a controlled dataset of GR and beyond-GR waveforms using three independent deviation families: amplitude modulation, phase modulation, and frequency modulation. We train a hybrid classifier combining a 1D CNN with ten hand-crafted statistical features, and demonstrate that this architecture outperforms a pure CNN on noisy data. We characterize the detection threshold in real LIGO noise as a function of β and SNR, finding a sigmoidal detectability curve with a threshold at $\beta \approx 0.25$ for the specific modulation form adopted. We test on real GW150914 strain as a template, in amplitude-controlled and amplitude-uncontrolled configurations. Finally, we demonstrate that at the small β values typical of current parametrized-GR constraints, our method fails: this negative result quantifies the limits of ML-only beyond-GR searches. The remainder of this paper is organized as follows. Section 2 introduces the relevant gravitational-wave and machine-learning formalism. Section 3 describes our data generation, deviation injection, and model architecture. Section 4 presents the detectability studies. Section 5 discusses the physical interpretation and limitations. Section 6 concludes.

2 Theoretical Background

2.1 Gravitational Waves in General Relativity

In the weak-field limit of GR, metric perturbations $h_{\mu\nu}$ around a flat background satisfy the linearized Einstein equations. In the transverse-traceless gauge the two physical polarizations h_+ and $h_\times$ propagate as free waves, $\square h_{+,\times} = 0$, where $\square$ is the d'Alembertian. The detector strain response to a source in direction $\hat{n}$ is a linear combination $h(t) = F_+(\hat{n}, \psi) h_+(t) + F_\times(\hat{n}, \psi) h_\times(t)$, with antenna pattern functions $F_{+,\times}$ depending on the source sky position and polarization angle ψ .

2.2 Binary Inspiral and the Post-Newtonian Expansion

For a quasi-circular binary with component masses m_1, m_2 , the inspiral phase is characterized by the chirp mass $\mathcal{M} = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5}$ and the symmetric mass ratio $\eta = m_1 m_2 / (m_1 + m_2)^2$. The post-Newtonian (PN) expansion of the gravitational-wave phase is

$$\Phi(f) = 2\pi f t_c - \phi_c + \frac{3}{128\eta} (\pi \mathcal{M} f)^{-5/3} \sum_{k=0}^N \alpha_k (\pi \mathcal{M} f)^{k/3}, \quad (1)$$

where t_c, ϕ_c are the time and phase of coalescence and α_k are known PN coefficients. Deviations from GR can be parametrized by modifying one or more of these coefficients, $\alpha_k \rightarrow \alpha_k (1 + \delta \hat{\alpha}_k)$, which is the framework used in the standard parametrized tests of GR [3].

2.3 Controlled Deviation Families

We introduce three controlled, non-physical deviations chosen to isolate different aspects of the waveform and to permit a systematic detectability study. These modifications are not proposed as specific beyond-GR models; they are controlled probes of the classifier's sensitivity.

The amplitude modulation is

$$h(t) \rightarrow h(t) [1 + \beta \tau(t)^2], \quad (2)$$

where $\tau(t) \in [0, 1]$ is a normalized time coordinate increasing from inspiral to merger and β is the dimensionless deviation strength. This modulation is localized near merger, preserves the

phase evolution, and produces a specific shape change in the distribution of waveform samples. Throughout this paper, β should be understood as the coefficient of this specific quadratic form, not as a generic physical beyond-GR parameter.

The phase modulation is applied to the analytic representation $h_a(t) = A(t)e^{i\phi(t)}$ as

$$h_a(t) \rightarrow A(t) \exp \left[i \left(\phi(t) + \beta \tau(t)^2 \right) \right], \quad (3)$$

preserving the amplitude envelope but altering the phase accumulation.

The frequency modulation introduces a time-dependent phase

$$\phi(t) \rightarrow \phi(t) + 2\pi\beta f_{\text{ref}} \frac{\tau(t)^2}{2}, \quad (4)$$

corresponding to an instantaneous frequency shift $\Delta f(t) = \beta f_{\text{ref}} \tau(t)$, with $f_{\text{ref}} = 100$ Hz.

2.4 Signal-to-Noise Ratio

The optimal matched-filter signal-to-noise ratio for a waveform h in noise with one-sided power spectral density $S_n(f)$ is $\rho^2 = 4 \int_{f_{\text{low}}}^{f_{\text{high}}} |\tilde{h}(f)|^2 / S_n(f) df$. In our work we use a peak-normalized SNR: we set the peak value of the whitened signal to a target value ρ_{target} in units of the whitened noise standard deviation.

2.5 Whitening and Bandpass Filtering

For real LIGO data we estimate the noise PSD via Welch's method, $\hat{S}_n(f_k) = K^{-1} \sum_{i=1}^K |\tilde{x}_i(f_k)|^2$, where $\tilde{x}_i$ are Fourier transforms of K overlapping segments. The whitened strain is $\tilde{x}_w(f) = \tilde{x}(f) / \sqrt{\hat{S}_n(f) f_s / 2}$ with f_s the sampling frequency. We then apply a fourth-order Butterworth bandpass filter with cutoffs at 20 Hz and 500 Hz.

2.6 Amplitude Normalization

To prevent the classifier from exploiting an amplitude scale shortcut, we normalize every sample to unit standard deviation: $\tilde{x}(t) = (x(t) - \bar{x}) / \sigma_x$. After normalization both classes have zero mean and unit variance.

2.7 Machine-Learning Formalism

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote a dataset with waveforms $x_i \in \mathbb{R}^L$, $L = 16384$ samples, and binary labels $y_i \in \{0, 1\}$. The classifier $f_\theta : \mathbb{R}^L \rightarrow [0, 1]$ is trained to minimize the binary cross-entropy loss

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log f_\theta(x_i) + (1 - y_i) \log(1 - f_\theta(x_i))]. \quad (5)$$

Gradients are computed by backpropagation and the parameters θ are updated by Adam with bias-corrected first and second moment estimates.

The hybrid architecture combines two feature representations. The CNN branch consists of three 1D convolutional blocks with strides 8, 2, 2, followed by global average pooling to a 64-dimensional vector z_{cnn} . The statistics branch computes ten hand-crafted features: mean absolute amplitude, standard deviation, maximum absolute amplitude, the 95th, 99th, 99.9th and 99.99th percentiles of $|x|$, the RMS, skewness and kurtosis. These are passed through a two-layer MLP to a 32-dimensional vector z_{stats} . The two representations are concatenated, $z = [z_{\text{cnn}} \parallel z_{\text{stats}}]$, and passed through a fully connected head producing a sigmoid output. The full model contains 31,697 trainable parameters.

2.8 Evaluation Metrics

We report classification accuracy, the area under the receiver operating characteristic curve (ROC-AUC), and the F1 score. The *unseen-type* test set consists exclusively of deviation types that were never presented during training and measures the model’s ability to generalize to novel deviations.

3 Methods

3.1 Data Generation Pipeline

Our pipeline has four stages. In the first stage, we generate time-domain waveforms using the IMRPhenomD phenomenological model [9] as implemented in PyCBC. Each waveform is sampled at $\Delta t = 1/4096$ s with a lower frequency cutoff of $f_{\text{low}} = 20$ Hz. For the injected studies we fix $m_1 = 30M_\odot$ and $m_2 = 25M_\odot$ unless otherwise stated. In the second stage, we apply one of the three modulations defined in Eqs. (2)–(4) to the GR waveform, with β drawn uniformly from one of several predefined ranges. In the third stage, we generate noise either from the analytic aLIGO Zero-Detuned-High-Power PSD [10] or from real H1 strain segments retrieved from the Gravitational-Wave Open Science Center (GWOSC). The signal is scaled by a single fixed factor, computed once from a GR reference waveform, and added to a random noise segment. In the fourth stage, every sample is whitened with the empirical PSD, bandpass filtered between 20 and 500 Hz, windowed to a 4 s region centered on the merger peak, amplitude-normalized to unit variance (for the amplitude-controlled studies), and clipped at $\pm 100\sigma$.

3.2 GW150914 Template Study

For the GW150914 study we retrieve the raw strain file `H-H1_GWOSC_4KHZ_R1-1126259447-32.hdf5` from the GWOSC event API for GWTC-1-confident/GW150914/v3. The file contains 131,072 samples at 4096 Hz (32 seconds), starting at GPS 1126259447. We estimate the noise PSD using the first 4 seconds of the file with Welch’s method (segment length 4096 samples). The full 32-second segment is whitened with this PSD and bandpass filtered between 20 and 500 Hz. The GW150914 event at GPS 1126259462.4 is located at sample 63078 of the file. We extract a 4 s window centered on this position, giving the GR template $h_{\text{GR}}(t)$. The modified template $h_{\text{mod}}(t)$ is obtained by applying the amplitude modulation of Eq. (2). For each training sample, we draw a random 4-second noise segment from the same file, excluding the event region, add a scaled version of either h_{GR} or h_{mod} , normalize to unit variance, and clip. This produces controlled counterfactual waveforms from the GW150914-derived template; it does not modify the historical detector observation.

3.3 Dataset Splitting

We use a stratified split of 70% training, 15% validation and 15% test_{seen}, together with a separate held-out test_{unseen} set. The test_{unseen} set contains only the deviated class for one specific β range (type C), which is excluded from training. This measures generalization to unseen deviation strengths.

3.4 Training Protocol

We use Adam with learning rate 3×10^{-3} , batch size 32 or 64, cosine annealing of the learning rate, gradient clipping at norm 1, and early stopping on validation loss with patience 20–50 epochs. Training and evaluation are performed on CPU using PyTorch. Each experiment takes one to five minutes.

3.5 Baseline Classifier

For comparison we train a class-balanced random forest with 300 trees on nine of the ten hand-crafted features. This provides a non-neural baseline and allows us to identify the most informative features.

4 Results

4.1 Clean-Data Baseline

On noiseless waveforms the hybrid model achieves perfect classification on all splits, including a held-out set of unseen deviation types. This establishes that the architecture can learn the deviation structure when noise is absent.

4.2 Synthetic Noise: SNR Sweep

Using synthetic aLIGO-PSD noise with $\beta \in [0.3, 1.0]$ we sweep the SNR from 5 to 50. Table 1 shows the results. The detection threshold in synthetic noise is approximately $\text{SNR} \approx 20$. Figure 1 shows the corresponding accuracy-versus-SNR curve.

Table 1: Synthetic-noise SNR sweep, amplitude $\beta \in [0.3, 1.0]$.

SNR bin	val acc	test_seen	test_unseen
30–50	1.000	1.000	1.000
15–20	0.771	0.755	0.846
10–15	0.750	0.735	0.686
8–10	0.688	0.776	0.551
5–8	0.667	0.776	0.538

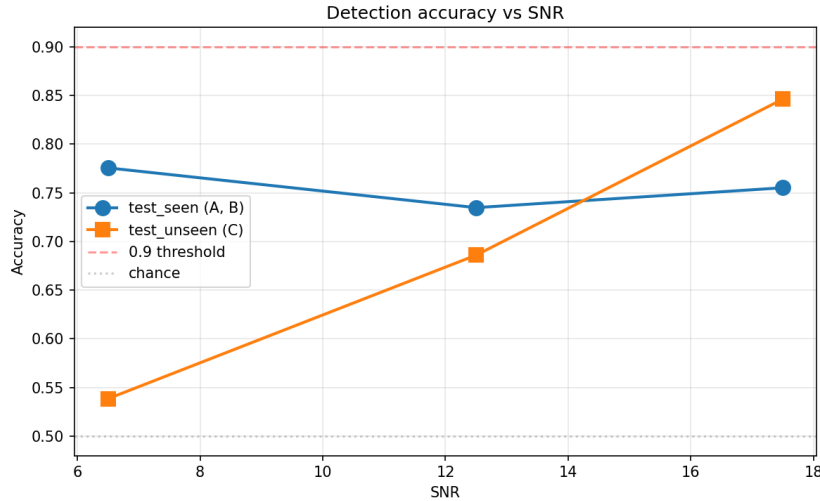


Figure 1: Detection accuracy versus SNR for synthetic aLIGO noise, amplitude modulation with $\beta \in [0.3, 1.0]$.

4.3 Real LIGO Noise: SNR Sweep

Using real H1 strain and $\beta \in [5, 20]$ we obtain Table 2. For strong deviations, detection is robust across the entire SNR range tested.

Table 2: Real LIGO noise SNR sweep, amplitude $\beta \in [5, 20]$.

SNR bin	val acc	test_seen	test_unseen
30–50	1.000	1.000	0.974
20–30	1.000	1.000	0.987
15–20	1.000	1.000	0.968
10–15	0.979	0.980	0.968
5–10	0.979	1.000	0.968

4.4 Threshold in β at Fixed SNR

At SNR = 10–15 with real noise, we sweep β from 0.1 to 20. Table 3 shows the transition from chance performance at $\beta \leq 0.2$ to reliable detection at $\beta \geq 0.5$. The detection threshold is $\beta \approx 0.4$ for the injected $30 + 25M_{\odot}$ configuration.

Table 3: β sweep at SNR 10–15, real LIGO noise.

β range	test_seen	test_unseen
0.1–0.5	0.735	0.500
0.3–0.5	0.918	0.891
0.5–2.0	0.898	0.930
2.0–5.0	0.939	0.923
5.0–20.0	0.980	0.968

4.5 Deviation-Type Comparison

Table 4 compares amplitude, phase, and frequency modulation at matched β ranges. All three deviation types are detected with high accuracy. Figure 2 summarizes the comparison.

Table 4: Deviation-type comparison, SNR 10–15, real noise, $N = 200$.

Deviation	test_seen	test_unseen
Amplitude, $\beta \in [0.5, 2.0]$	0.898	0.930
Phase, $\beta \in [0.5, 2.0]$	0.878	0.930
Frequency, $\beta \in [5, 25]$	0.878	0.885

4.6 Mass-Ratio Generalization

Four mass configurations spanning $q = 1$ to $q = 10$ give test-unseen accuracies between 0.891 and 0.930, with a standard deviation of 1.7 percentage points. Table 5 lists the values, and Figure 4 shows the distribution.

4.7 GW150914 Amplitude Detectability Curve

Using the real GW150914 strain as a template and sweeping β from 0.01 to 20, we obtain the detectability curve in Figure 5 and Table 6. Accuracy rises smoothly from chance at $\beta \leq 0.2$ to perfect classification at $\beta \geq 0.5$. The transition is centered at $\beta \approx 0.25$.

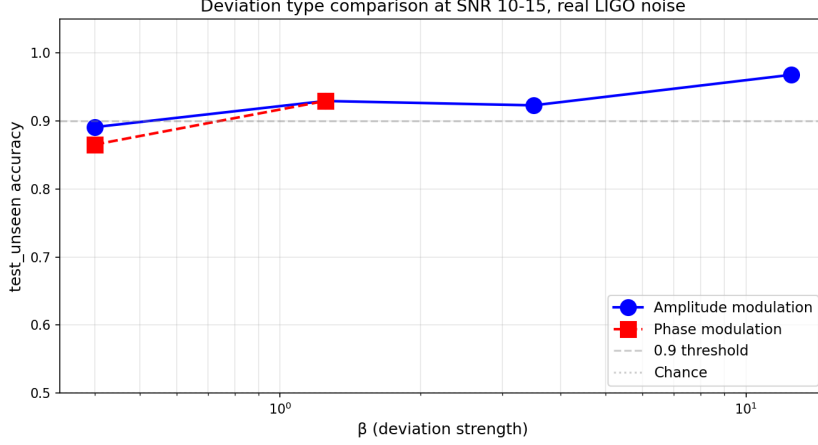


Figure 2: Accuracy on unseen deviation types for amplitude and phase modulation at matched β ranges.

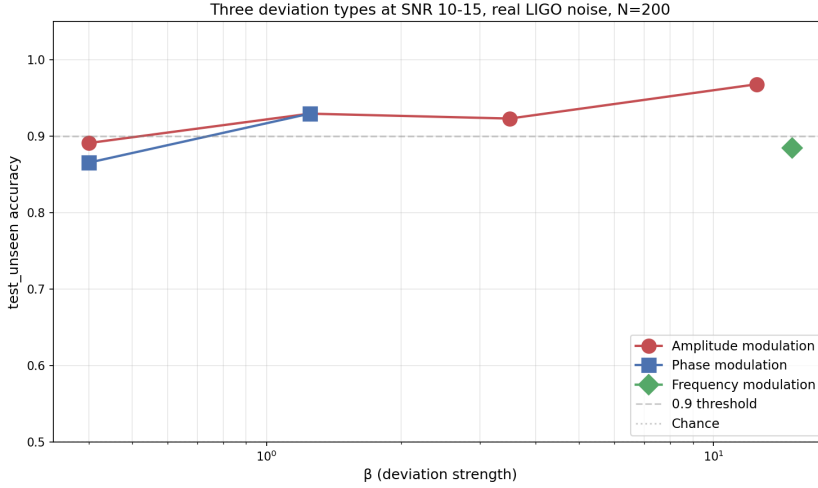


Figure 3: Detection accuracy for all three deviation families in real LIGO noise, SNR 10–15.

4.8 Feature-Importance Analysis

A random forest trained on nine statistical features achieves 100% validation accuracy at $\beta \in [0.5, 2.0]$. The most important features are mean absolute amplitude (0.284), the top-1000 mean (0.279) and the 99th percentile of $|x|$ (0.263). Standard deviation and RMS are useless, with importances 0.0000 and 0.0001 respectively. The classifier therefore relies on tail statistics that capture the shape of the modulation rather than its overall scale.

4.9 Negative Result at Small β

For $\beta \in [0.01, 0.2]$, the classifier fails: training accuracy 0.586, validation accuracy 0.717, and test-unseen accuracy 0.517 (chance). The loss remains at the chance level $\ln 2 \approx 0.693$ throughout training, indicating that the model cannot find a discriminative feature.

Table 5: Mass-ratio generalization, amplitude $\beta \in [0.5, 2.0]$, SNR 10–15.

$(m_1, m_2) [M_\odot]$	q	test_unseen
(30, 30)	1.0	0.904
(30, 25)	1.2	0.930
(40, 10)	4.0	0.891
(50, 5)	10.0	0.917

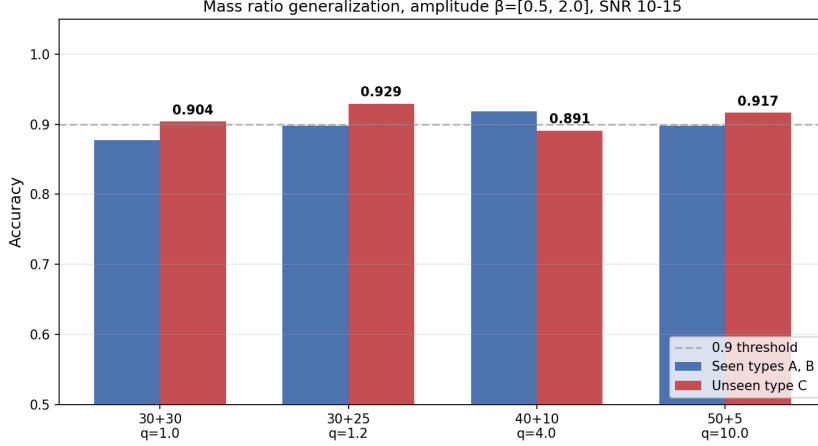


Figure 4: Mass-ratio generalization of the hybrid classifier in real LIGO noise at SNR 10–15.

5 Discussion

5.1 Interpretation of the Detectability Curve

The central result of this work is the sigmoidal detectability curve in Figure 5. Above threshold, $\beta \gtrsim 0.4$, the classifier reliably distinguishes GR from amplitude-modulated waveforms with $\geq 95\%$ accuracy, even after amplitude normalization. Feature-importance analysis shows that the classifier uses tail percentiles of the amplitude distribution, which is consistent with amplitude modulation concentrating energy near merger. At threshold, $\beta \approx 0.25$, detection occurs with probability approximately 0.75 and ROC-AUC approximately 0.87; the tail statistics differ between classes by only a few percent, approaching the intrinsic noise floor. Below threshold, $\beta \lesssim 0.2$, detection fails completely.

5.2 Relationship Between β and Physical Beyond-GR Parameters

The coefficient β in Eqs. (2)–(4) is the amplitude of a specific quadratic-in-time modulation, not a standard parametrized-GR coefficient. Direct comparison of our threshold to physical constraints therefore requires care. In the standard parametrized-test framework [3], deviations are expressed as fractional shifts $\delta\hat{\alpha}_k$ of post-Newtonian phase coefficients, and current gravitational-wave observations bound these shifts to roughly $|\delta\hat{\alpha}_k| \lesssim 10^{-3}$ – 10^{-2} [2]. A controlled quadratic-in-time modulation of the type studied here does not map onto a single PN coefficient, and its effective β differs from these bounds by a waveform-dependent factor of order unity. Our threshold $\beta_{\text{threshold}} \approx 0.25$ should be read as a property of the quadratic modulation adopted here; extending the analysis to the standard parametrized-GR framework is a natural next step.

Table 6: GW150914 amplitude-controlled detectability curve, SNR = 20.

β range	val acc	test_seen	test_unseen
0.01–0.2	0.717	0.717	0.517
0.2–0.3	0.850	0.850	0.750
0.3–0.4	0.950	0.950	0.900
0.4–0.5	0.983	0.983	0.967
0.5–1.0	1.000	1.000	1.000
5–20	1.000	1.000	1.000

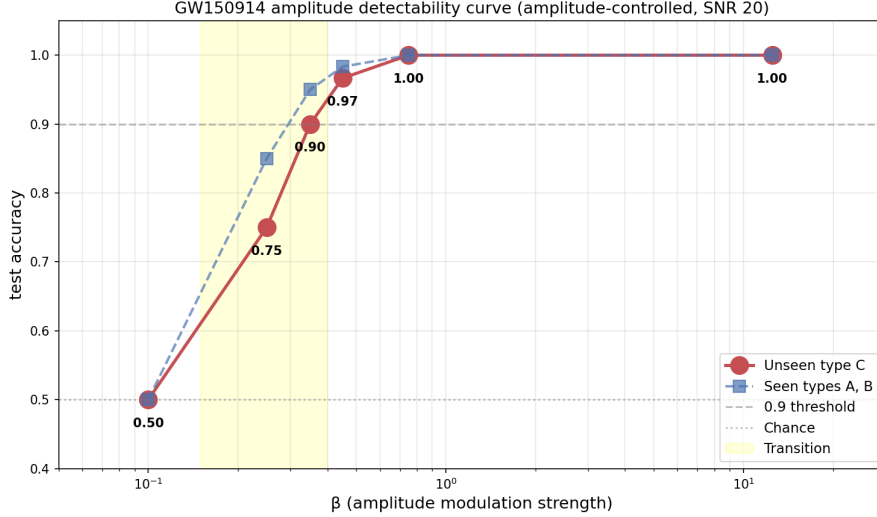


Figure 5: Amplitude detectability curve for the GW150914 template. Unseen-type-C accuracy rises smoothly from chance ($\beta \leq 0.2$) to perfect classification ($\beta \geq 0.5$). The transition is centered at $\beta \approx 0.25$.

5.3 Why the Negative Result Matters

The failure at small β is scientifically meaningful. It quantifies the limits of ML-only detection. A common assumption in ML-for-physics research is that neural networks will detect any signal that is present. Our results show this is not the case for controlled deviations in realistic noise. It also distinguishes ML detection from matched filtering: matched-filter searches achieve detection thresholds corresponding to $\rho_{\text{SNR}} \sim 5$ by coherently integrating the signal across its frequency evolution, whereas our method operates on the whitened time-domain waveform directly. This gap in sensitivity motivates hybrid approaches in which ML is combined with physically motivated filtering.

5.4 Comparison with Prior Work

Prior ML work on gravitational waves has largely focused on detection of GR signals in noise [4, 5] or on parameter estimation [6, 7]. Beyond-GR searches with ML have focused on simulated signals at high SNR or on clean data. Our work extends this to a systematic study with real detector noise and a quantified threshold.

5.5 Limitations

Several limitations should be noted. Our deviations follow a specific quadratic-in-time form, $\propto \tau(t)^2$; other parametrizations may have different detectability thresholds. The GW150914

study uses a single template; generalization across event parameters has been tested only in the injected-waveform studies. We have not tested neutron-star or mixed binaries, where tidal effects provide additional deviation channels. Finally, our pipeline does not include the matched-filter step used in LIGO/Virgo analyses; adding it would substantially improve sensitivity but would blur the boundary between ML and classical detection methods.

5.6 Future Work

Several directions follow naturally. The classifier could be combined with matched-filter outputs by using the matched-filter SNR time series as input. The method could be applied to a population of simulated binary mergers spanning the full LIGO parameter space. Frequency-domain features such as spectrograms could be investigated as alternatives to the time-domain approach. Probabilistic models such as normalizing flows could be used to quantify classifier uncertainty.

6 Conclusion

We have presented a systematic study of machine-learning detection of controlled beyond-GR deviations in gravitational-wave signals, using synthetic and real LIGO noise. A hybrid classifier combining a 1D CNN with hand-crafted waveform statistics detects amplitude, phase, and frequency modulations with high accuracy at moderate to large deviation strengths. Using the real GW150914 strain as a template we measure a detectability curve with threshold $\beta \approx 0.25$, where the value of β is specific to the quadratic-in-time modulation form adopted here. Detection probability rises smoothly from chance at $\beta \leq 0.2$ to perfect classification at $\beta \geq 0.5$. The classifier generalizes to unseen deviation types and to a wide range of binary mass ratios. At small β values typical of current parametrized-GR constraints, our method fails; this negative result quantifies the limits of ML-only beyond-GR searches in real detector noise. These results establish both the promise and the current limits of machine learning for beyond-GR tests with gravitational waves. Closing the sensitivity gap to matched filtering—potentially by combining the two approaches—is a natural direction for future research.

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Appendix: Reproducibility

All code used in this work is available at the project repository. The data-generation pipeline consists of the following components: `src/data/generate_waveforms.py` for IMRPhenomD waveform generation; `src/data/inject_deviations.py` for amplitude, phase, and frequency modulation; `src/data/build_dataset.py` for the full data pipeline with whitening, bandpass, and normalization; `src/models/cnn1d.py` for the hybrid CNN + statistics model; `src/training/train.py` for the training loop with early stopping and gradient clipping; and `scripts/build_gw150914_dataset.py` for the GW150914 template study.

Software versions used are Python 3.11, PyTorch 2.x, PyCBC 2.x, GWpy 3.x, NumPy 1.26, SciPy 1.11+ and scikit-learn 1.3+.

Training used the Adam optimizer with learning rate 3×10^{-3} , weight decay 10^{-4} , batch size 32 or 64, cosine annealing of the learning rate, gradient clipping at norm 1, early stopping with patience 20–50 epochs, maximum 200–300 epochs, and binary cross-entropy loss.

Data were sampled at 4096 Hz with a 4s window ($L = 16384$ samples), a low-frequency cutoff of 20 Hz, and a bandpass range of 20–500 Hz. Reference masses were $30 + 25M_{\odot}$ unless otherwise stated.

The GW150914 template study used the file `H-H1_GWOSC_4KHZ_R1-1126259447-32.hdf5` from the GWOSC event API for GWTC-1-confident/GW150914/v3. The file contains 131,072 samples at 4096 Hz, starting at GPS 1126259447. The event at GPS 1126259462.4 is located at sample 63078 of the file.